The current project

on

Smart Dustbin(s1)

has been successfully accomplished in 1st semester of the degree of Bachelor of Technology for the course Workshop and Manufacturing Practices (PME 151)

by

Name of Team Members					
S. No	**Name**	**Class Roll No.**	**University Roll No.**	**Section**	**Passport Size (Image)**
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Smart Dustbin(s1)

1. Introduction related to Project

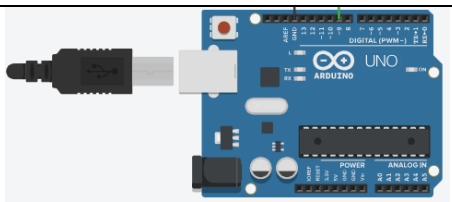
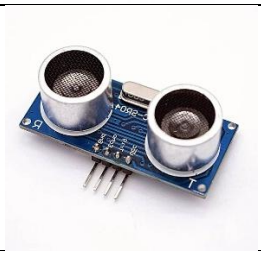
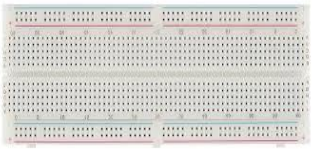
A **smart dustbin** is a modern waste container designed to make waste disposal cleaner, easier, and more efficient. Unlike traditional bins, a smart dustbin uses sensors to detect when someone is near and opens its lid automatically. This reduces the need to touch the bin, helping maintain better hygiene—especially in public places, schools, hospitals, and homes.

Many smart dustbins can also monitor how full they are. When the bin reaches a certain level, it can send a notification to workers or a connected device, ensuring waste is collected on time and preventing overflow. This makes waste management more organized and reduces bad odors and litter.

Some advanced smart dustbins even support basic waste segregation, helping people dispose of waste more responsibly. Overall, smart dustbins make everyday waste disposal more convenient while supporting cleaner surroundings and smarter city systems.

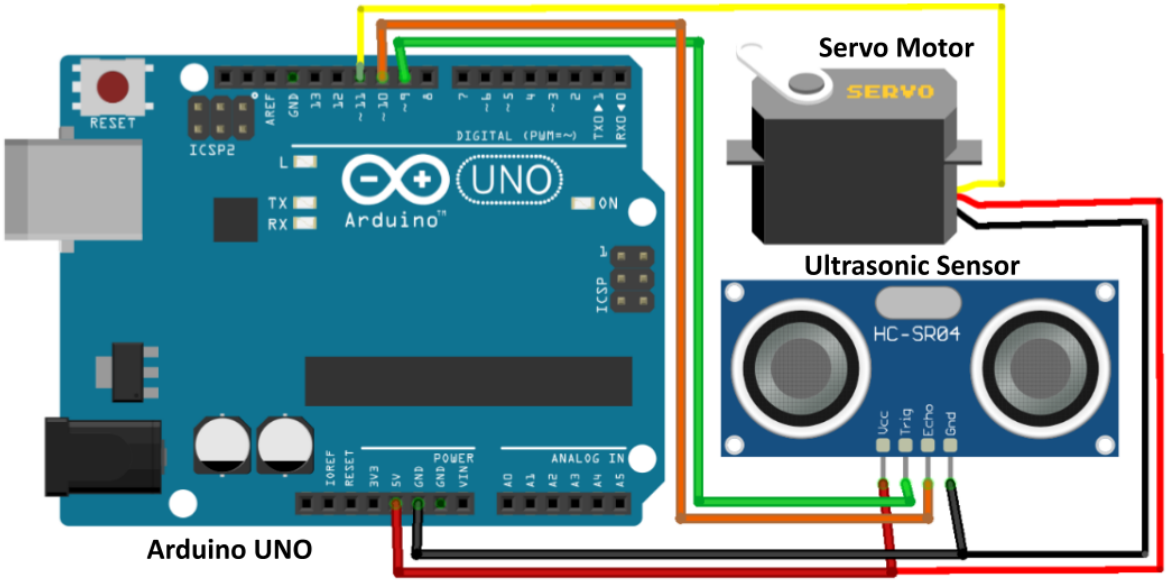
Smart Dustbin(s1)

2. List of Component Required

S. No	Name of Component	Specification	Quantity	Image
1	Arduino Uno	ATmega328P	1	
2	Jumper wires	Male -Male	20	
3	Jumper wires	Male -Female	20	
4	Servo motor	SG90	1	
5	Ultrasonic sensor	HC-SR04	1	
6	breadboard		1	
7	Dustbin		1	

Smart Dustbin(s1)

1. Schematic Diagram (Connection Diagram)



Smart Dustbin(s1)

2. Steps of Construction (Complete process of making)

- **Fix the ultrasonic sensor**

Attach the sensor at the front or top of the bin. This sensor will sense a hand or object coming near.

- **Set up the microcontroller**

Use an Arduino. This will act like the "brain" of the dustbin and control the motor based on the sensor reading.

- **Attach the servo motor**

Fix the servo motor to the lid of the dustbin. The motor will help the lid open and close automatically.

- **Do proper wiring**

Connect the ultrasonic sensor and servo motor to the Arduino using jumper wires. Make sure the power and ground wires are connected correctly.

- **Write and upload the code**

Write a simple program that says:
– If something comes near → open lid
– After few seconds → close lid

Here is the code file link:-
https://www.youtube.com/redirect?event=video_description&redir_token=QUFFLUhqbC01c0NGb0ZOaVlRY0htMnNXSmltcy0tWHRvZ3xBQ3Jtc0ttendHcDNveEFFV1gzMm13djY4YXlKTEVXaWRsQ3ZTVktWVkZ5aEVFTWxzeF9CdEJDZDNwNzNtTHlJR3VOb2JFalN1SHEySFZmY1NpWUF2R2hpZDBJNUpSR3RTRnFjOWQ0UzFpNllZWjJJVE95Njlvcw&q=https%3A%2F%2Froboarmypublicfiles.s3.ap-south-1.amazonaws.com%2Fcode%2Bdoc%2FA.H.docx&v=whoqckyTrw0
Upload this program to the Arduino.

- **Test the working**

Bring your hand near the sensor and check if the lid opens and closes properly. Adjust the sensor position if needed.

- **Finish the setup**

Fix all components securely inside or around the dustbin, cover wires neatly, and ensure the lid moves smoothly.

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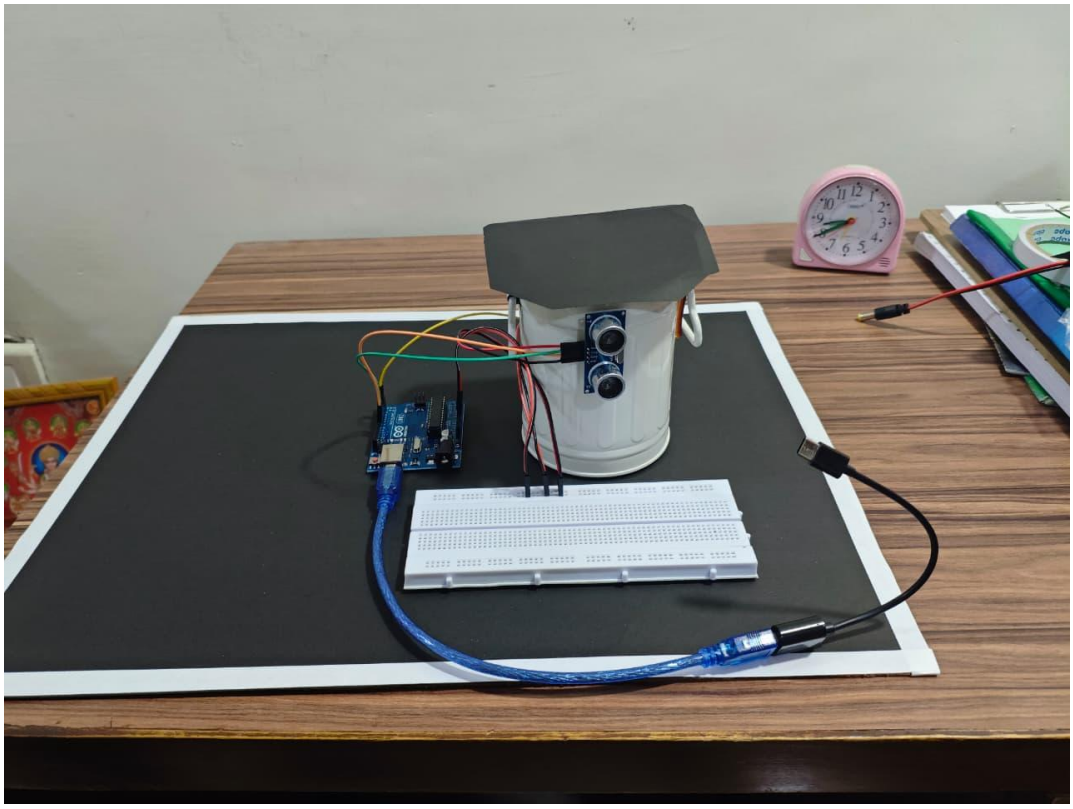
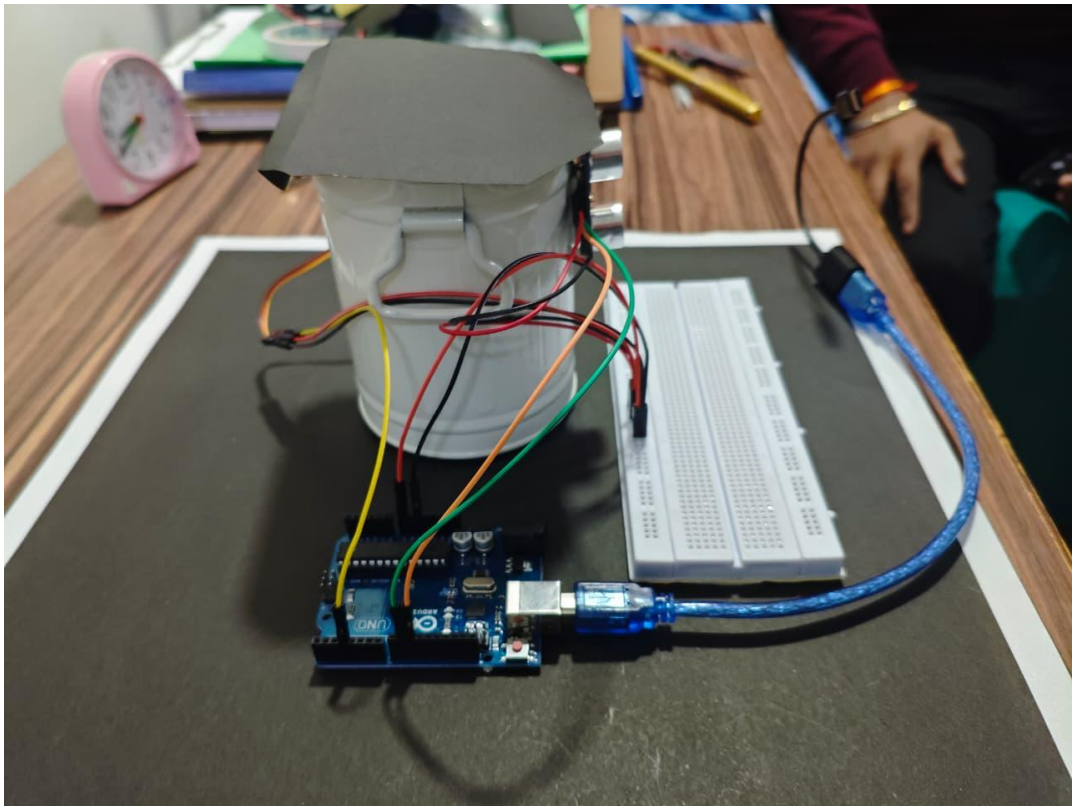
3. Original Image of Final Product





Smart Dustbin(s1)

4. **Uses and Application**

<u>Usage</u>

- **Automatically opens the lid when you come close.**
- **Helps you throw waste without touching the bin.**
- **Shows or alerts when it is full.**
- **Helps keep the area clean and hygienic.**

<u>Applications</u>

- **Homes: Makes waste disposal easy and germ-free.**
- **Schools & Colleges: Keeps campuses cleaner and teaches good habits.**
- **Hospitals: Reduces infection risk by avoiding touch.**
- **Offices: Maintains a clean workplace.**
- **Public Places: Prevents overflowing garbage and keeps the environment tidy.**

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5. Future Scope

- **Automatic waste sorting**
It will separate plastic, metal, paper, and wet waste on its own.

- **Connected to the internet**
It will send notifications when it is full, helping cleaners save time.

- **Better cleanliness in cities**
Smart bins can reduce overflowing garbage and keep streets cleaner.

- **Used in smart homes**
Houses will use smart bins to manage waste automatically.

- **Helpful for recycling**
Sorting waste properly will increase recycling and reduce pollution.

- **Energy saving**
Some smart bins may run on solar power.

- **Useful in public places**
Malls, airports, colleges, and parks will use smart bins for better waste management.

6. References/ links/ etc. related to component

https://youtu.be/whoqckyTrw0?si=3JSo8sQYIFqMcDj6